

Algorithm

1. Start.
2. Read the size **n** of the first array.
3. Check whether **n** is positive.
4. Read **n** elements into the array.
5. Display the menu:
 - Count even, odd, positive, negative, and zero elements.
 - Search for an element using linear search.
 - Merge two arrays.
 - Exit.
6. Read the user's choice.
7. Use a **switch** statement:
 - **Choice 1:** Traverse the array and count even, odd, positive, negative, and zero elements.
 - **Choice 2:** Read the element to search. Traverse the array using linear search and display its position if found.
 - **Choice 3:** Read the second array and combine the first and second arrays into a third array.
 - **Choice 4:** Exit.
 - Otherwise, display "Invalid choice".
8. Repeat the menu until the user selects Exit.
9. Stop.

Test Case Table

Test Case	Operation	Input	Expected Output
1	Count elements	5 → 2 -3 0 7 -4	Even = 3, Odd = 2, Positive = 2, Negative = 2, Zero = 1
2	Linear search	Array: 10 20 30 40 50, Search: 30	Element found at position 3
3	Linear search	Array: 10 20 30 40 50, Search: 60	Element not found
4	Merge	A: 1 2 3, B: 4 5 6	Merged: 1 2 3 4 5 6
5	Invalid	Choice = 7	Invalid choice
6	Exit	Choice = 4	Program terminated

Note: Zero is counted as an even number, but it is neither positive nor negative.

C Program

```
#include <stdio.h>
```

```
int main()  
{
```

```

int a[100], b[100], merged[200];
int n, n2;
int choice;
int i, search;
int even, odd, positive, negative, zero;
int found;

/* Input first array */
do
{
    printf("Enter the size of the array: ");
    scanf("%d", &n);

    if (n <= 0 || n > 100)
        printf("Invalid size. Enter a value between 1 and
100.\n");

} while (n <= 0 || n > 100);

printf("Enter %d elements:\n", n);

for (i = 0; i < n; i++)
{
    scanf("%d", &a[i]);
}

do
{
    printf("\n----- MENU ----- \n");
    printf("1. Count even, odd, positive, negative and
zero elements\n");
    printf("2. Search for an element using linear
search\n");
    printf("3. Merge two arrays\n");
    printf("4. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            even = 0;
            odd = 0;
            positive = 0;
            negative = 0;

```

```

zero = 0;

for (i = 0; i < n; i++)
{
    /* Count even and odd */
    if (a[i] % 2 == 0)
        even++;
    else
        odd++;

    /* Count positive, negative and zero */
    if (a[i] > 0)
        positive++;
    else if (a[i] < 0)
        negative++;
    else
        zero++;
}

printf("\nEven elements      = %d\n", even);
printf("Odd elements        = %d\n", odd);
printf("Positive elements    = %d\n", positive);
printf("Negative elements    = %d\n", negative);
printf("Zero elements       = %d\n", zero);

break;

```

case 2:

```

printf("Enter the element to search: ");
scanf("%d", &search);

```

```

found = 0;

```

```

for (i = 0; i < n; i++)
{

```

```

    if (a[i] == search)
    {

```

```

        printf("Element %d found at position
%d.\n",

```

```

            search, i + 1);

```

```

        found = 1;

```

```

        break;

```

```

    }

```

```

}

```

```

        if (found == 0)
        {
            printf("Element %d not found in the
array.\n",
                    search);
        }

        break;

    case 3:
        do
        {
            printf("Enter the size of the second
array: ");

            scanf("%d", &n2);

            if (n2 <= 0 || n2 > 100)
                printf("Invalid size. Enter a value
between 1 and 100.\n");

        } while (n2 <= 0 || n2 > 100);

        printf("Enter %d elements of the second
array:\n", n2);

        for (i = 0; i < n2; i++)
        {
            scanf("%d", &b[i]);
        }

        /* Copy first array */
        for (i = 0; i < n; i++)
        {
            merged[i] = a[i];
        }

        /* Copy second array */
        for (i = 0; i < n2; i++)
        {
            merged[n + i] = b[i];
        }

        printf("Merged array:\n");

        for (i = 0; i < n + n2; i++)

```

```
        {
            printf("%d ", merged[i]);
        }

        printf("\n");

        break;

    case 4:
        printf("Exiting the program...\n");
        break;

    default:
        printf("Invalid choice. Please enter 1 to
4.\n");
    }

    } while (choice != 4);

    return 0;
}
```

Execution

```
Enter the size of the array: 5
Enter 5 elements:
2 -3 0 7 -4

----- MENU -----
1. Count even, odd, positive, negative and zero elements
2. Search for an element using linear search
3. Merge two arrays
4. Exit
Enter your choice: 1

Even elements      = 3
Odd elements       = 2
Positive elements  = 2
Negative elements  = 2
Zero elements      = 1

----- MENU -----
1. Count even, odd, positive, negative and zero elements
2. Search for an element using linear search
3. Merge two arrays
4. Exit
Enter your choice: 2
Enter the element to search: 7
Element 7 found at position 4.

----- MENU -----
1. Count even, odd, positive, negative and zero elements
2. Search for an element using linear search
3. Merge two arrays
4. Exit
Enter your choice: 3
Enter the size of the second array: 3
Enter 3 elements of the second array:
10 20 30
Merged array:
2 -3 0 7 -4 10 20 30

----- MENU -----
1. Count even, odd, positive, negative and zero elements
2. Search for an element using linear search
3. Merge two arrays
4. Exit
Enter your choice: 4
Exiting the program...
bhavya@Bhavyas-MacBook-Air ~ %
```

This program uses **arrays**, **loops**, **switch-case**, **input validation**, **linear search**, and **array merging**, covering all four operations required for the Day-29 activity.